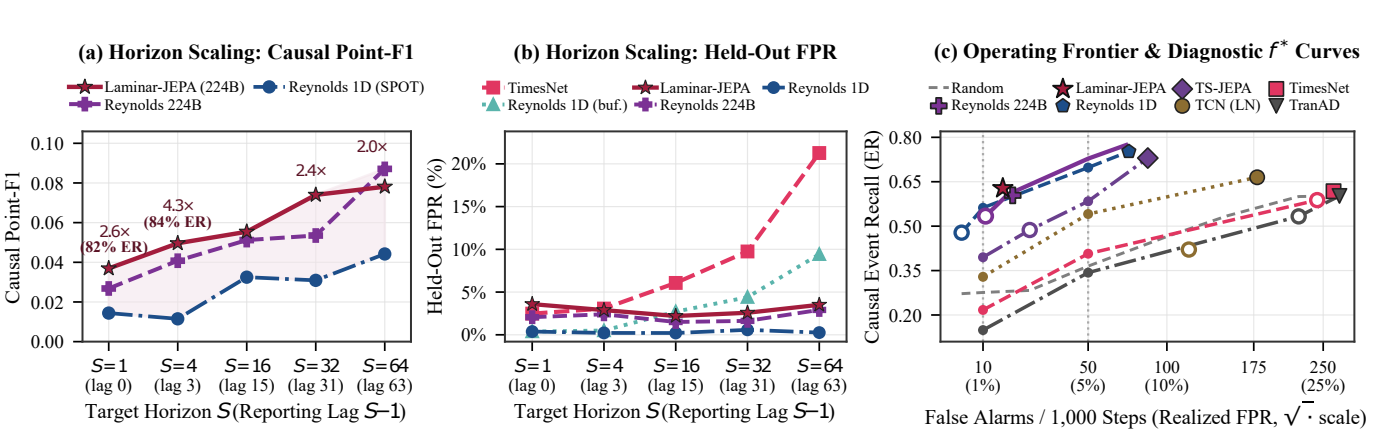

(a) Horizon Scaling: Causal Point-F1
Laminar-JEPA (224B)
Reynolds 1D (SPOT)
Reynolds 224B
Causal Point-F1
0.10
0.08
0.06
0.04
0.02
0.00
2.6×
(82% ER)
4.3×
(84% ER)
2.4×
2.0×
S=1
(lag 0)
S=4
(lag 3)
S=16
(lag 15)
S=32
(lag 31)
S=64
(lag 63)
Target Horizon S (Reporting Lag S−1)
(b) Horizon Scaling: Held-Out FPR
TimesNet
Laminar-JEPA
Reynolds 1D
Reynolds 1D (buf.)
Reynolds 224B
Held-Out FPR (%)
20%
15%
10%
5%
0%
S=1
(lag 0)
S=4
(lag 3)
S=16
(lag 15)
S=32
(lag 31)
S=64
(lag 63)
Target Horizon S (Reporting Lag S−1)
(c) Operating Frontier & Diagnostic f* Curves
Random
Reynolds 224B
Laminar-JEPA
Reynolds 1D
TS-JEPA
TCN (LN)
TimesNet
TranAD
Causal Event Recall (ER)
0.80
0.65
0.50
0.35
0.20
10
(1%)
50
(5%)
100
(10%)
175
250
(25%)
False Alarms / 1,000 Steps (Realized FPR, √· scale)